

Varianza della Loss Function nei NPQC

Studio del fenomeno tramite matrici non negative [1]

Falasca Matteo, Pian Sebastiano

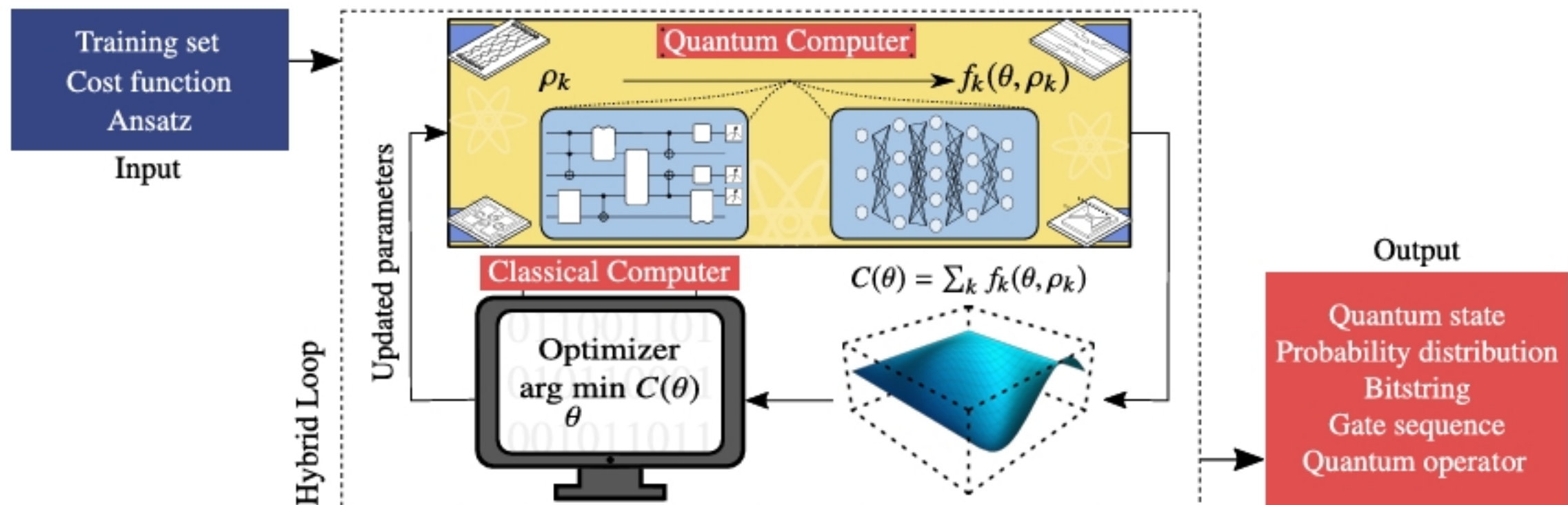
[1]: *Estimates of loss function concentration in noisy parametrized quantum circuits*

Parametrized Quantum Circuits

- I PQC si utilizzano in algoritmi ibridi quantistici-classici, che stanno alla base dei VQA
- Si basano sull'ottimizzazione di un parametro, massimizzando o minimizzando una funzione di Loss, tipicamente:

$$\mathcal{L}_{\rho,k} = \text{Tr}[\Phi_{\theta}(\rho)H]$$

2

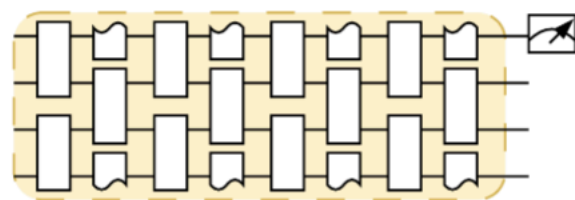


Barren Plateau

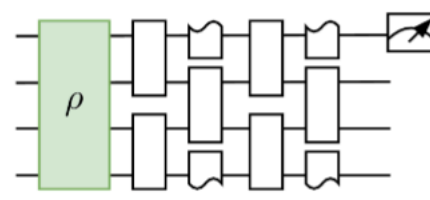
- Il fenomeno consiste nell'appiattimento esponenziale della Loss Function, che implica un aumento esponenziale del numero di interazioni richieste per l'ottimizzazione
- Cause: dimensionalità del sistema, entangling dello stato iniziale, entangling dell'osservabile e noise ambientale

Trainability – Theoretical Limitations

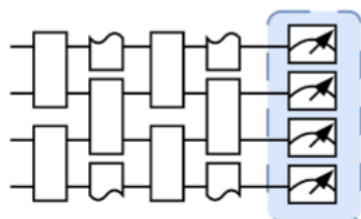
Exponential Decay in the loss function $\text{Var}[\ell_{\theta}(\rho, O)] \in \mathcal{O}\left(\frac{1}{b^n}\right)$



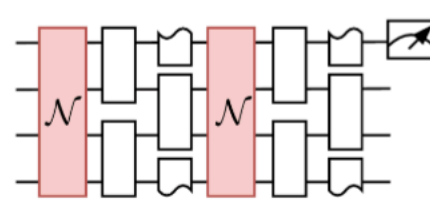
Circuit Expressivity



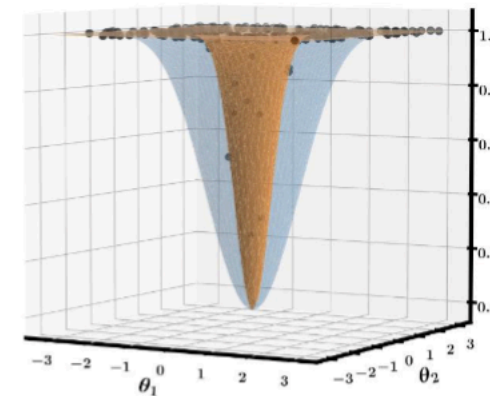
Entangled Initial State



Measurement Locality



Quantum Hardware Noise



Probabilistic
Barren plateau

Deterministic
Barren plateau

Vettori di Località

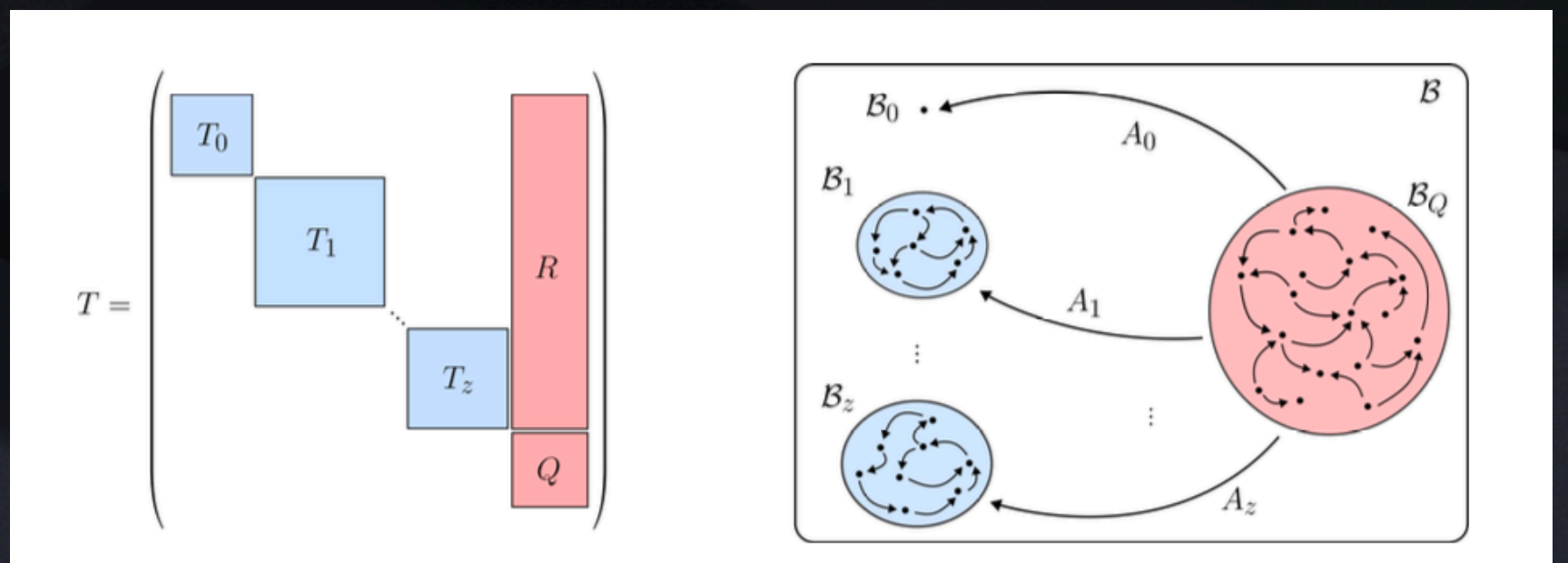
- Suddividendo lo spazio di Hilbert del sistema, suddividiamo lo spazio degli operatori limitati su H , $B = \bigoplus_{\kappa \in \{0,1\}^M} B_{\kappa}$ dove $k=0$ se gli operatori agiscono trivialmente su H_m
- Definiamo vettore di località: $(\ell_A)_\kappa = \sum_{j=1}^{d_\kappa} \text{Tr}[B_j A]^2$, dove B_j è un elemento di base di B_k

Matrice di Trasferimento

$$T_{\kappa,\lambda} = \frac{1}{d_\kappa} \sum_{j=1}^{d_\kappa} (\ell_{\Lambda(B_j)})_\lambda$$

$$\begin{bmatrix} T^{(1,1)} & 0 & \dots & T^{(1,n)} \\ 0 & T^{(2,2)} & \dots & T^{(2,n)} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & T^{(n,n)} \end{bmatrix}$$

- Contiene l'informazione sul trasferimento di località
- Può essere decomposta in una serie di blocchi irriducibili



Varianza analitica

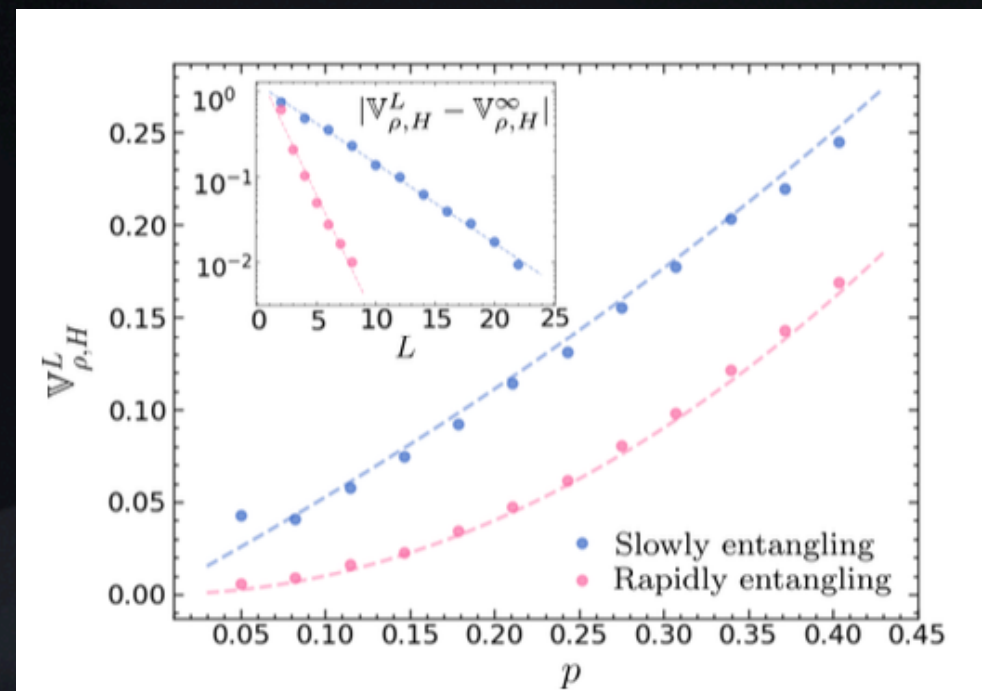
- $E_{\theta}(\text{Tr}[\Phi_{\theta}(\rho)H]^2) = (l_{\rho}, \prod_{l=1}^L T_l l_H) \rightarrow \mathbf{v}_{\rho,H}^L = (l_{\rho}, \prod_{l=1}^L T_l l_H) - \frac{\text{Tr}[H]^2}{d^2}$
- Se prendiamo un canale di noise del tipo:
 $\mathcal{E}_c(\rho) = (1-p)\mathcal{E}(\rho) + p\tilde{\rho}$, allora: $V_{\infty}^{\rho,H} = p^2(l_{\tilde{\rho}}(I - (1-p)^2 T)l_H)$ e
 se T è un proiettore, possiamo scomporre la varianza in due termini:
 - $V_{\infty}^{\rho,H} = (p/(2-p) - p^2)(l_{\tilde{\rho}}, Tl_H) + p^2(l_{\tilde{\rho}}, l_H)$
 - $|\frac{1}{L} \sum_{l=1}^L \mathcal{V}^l - \mathcal{V}^{\infty}| \simeq e^{-\beta L}$

Circuito e noise

- Stato iniziale: $|0\dots 0\rangle$
- Prendiamo una mappa di noise del tipo: $\mathcal{E}_c(\rho) = (1 - p)\mathcal{E}(\rho) + p\tilde{\rho}$

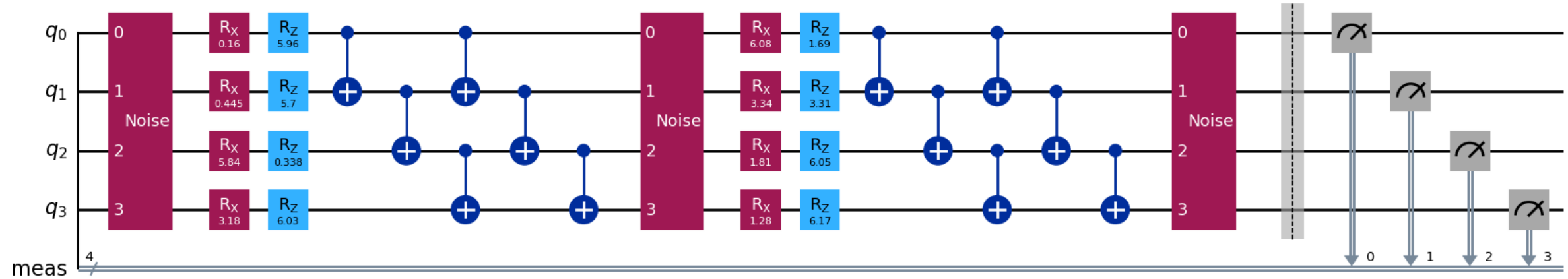
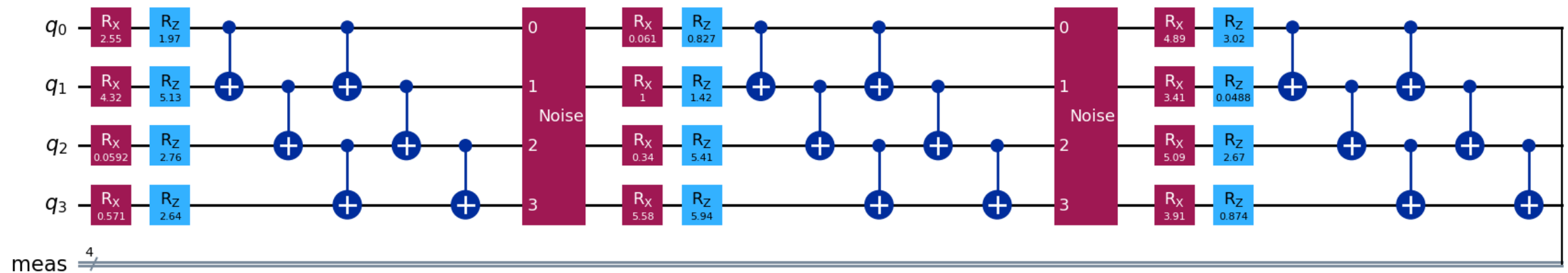
Con $\tilde{\rho} = \frac{(|0\rangle^{\otimes n} + |1\rangle^{\otimes n})(\langle 0|^{\otimes n} + \langle 1|^{\otimes n})}{2^n}$

$$H = h \sum_{k=1}^n 2^{n/2} Z_k \otimes Z_{k+1}$$



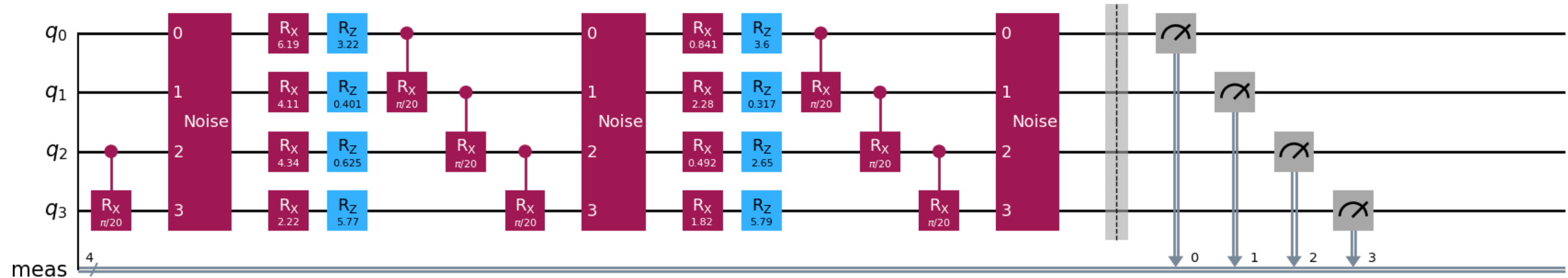
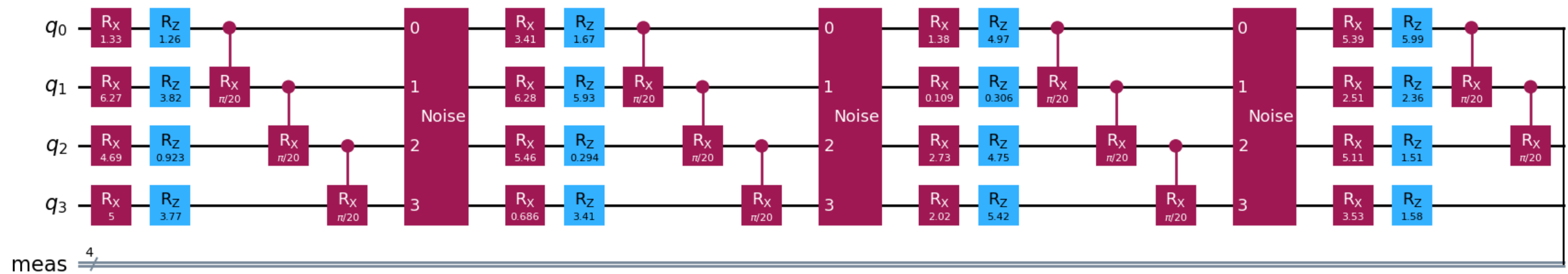
Circuiti considerati

Fast Entangling

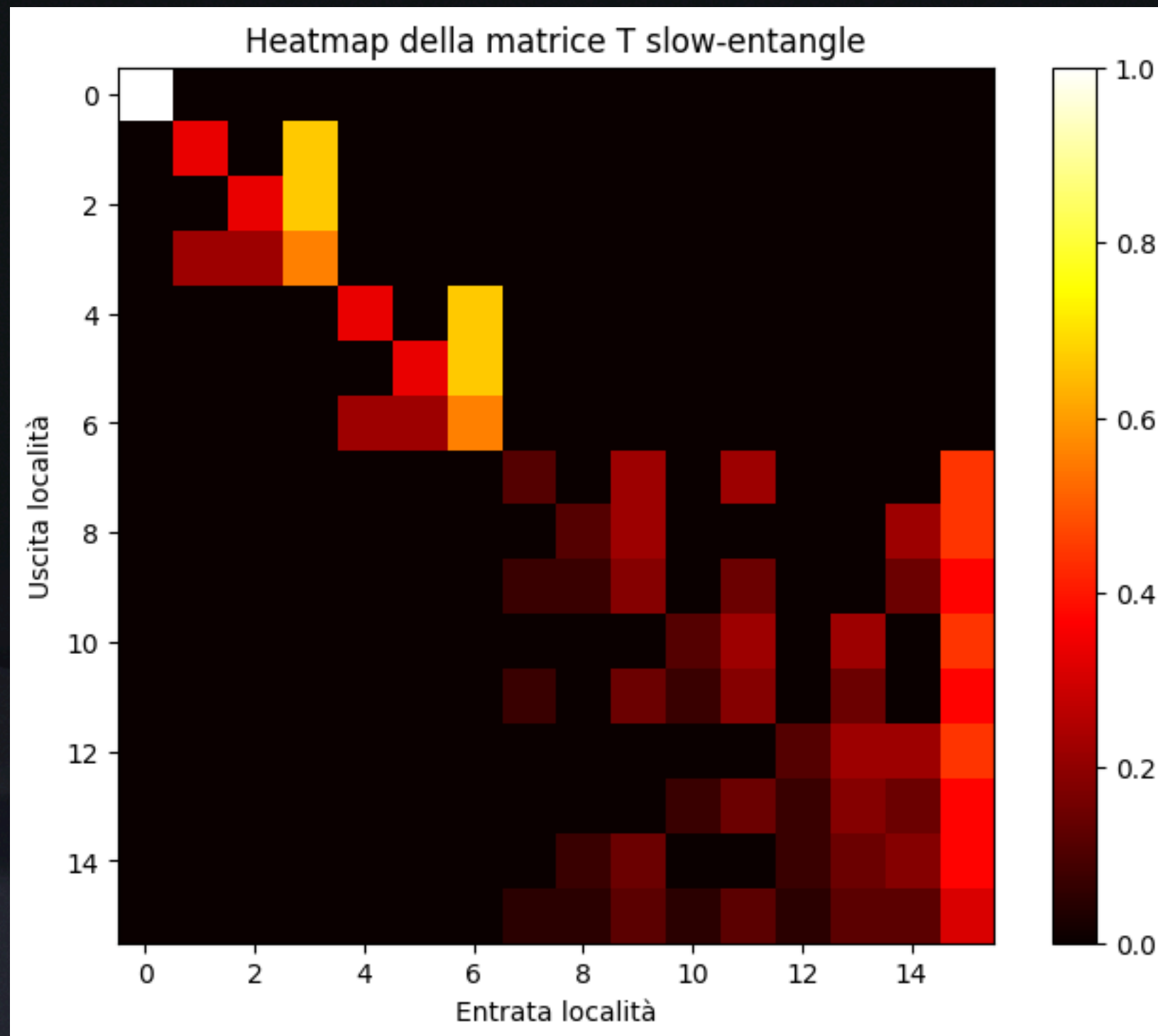


Circuiti considerati

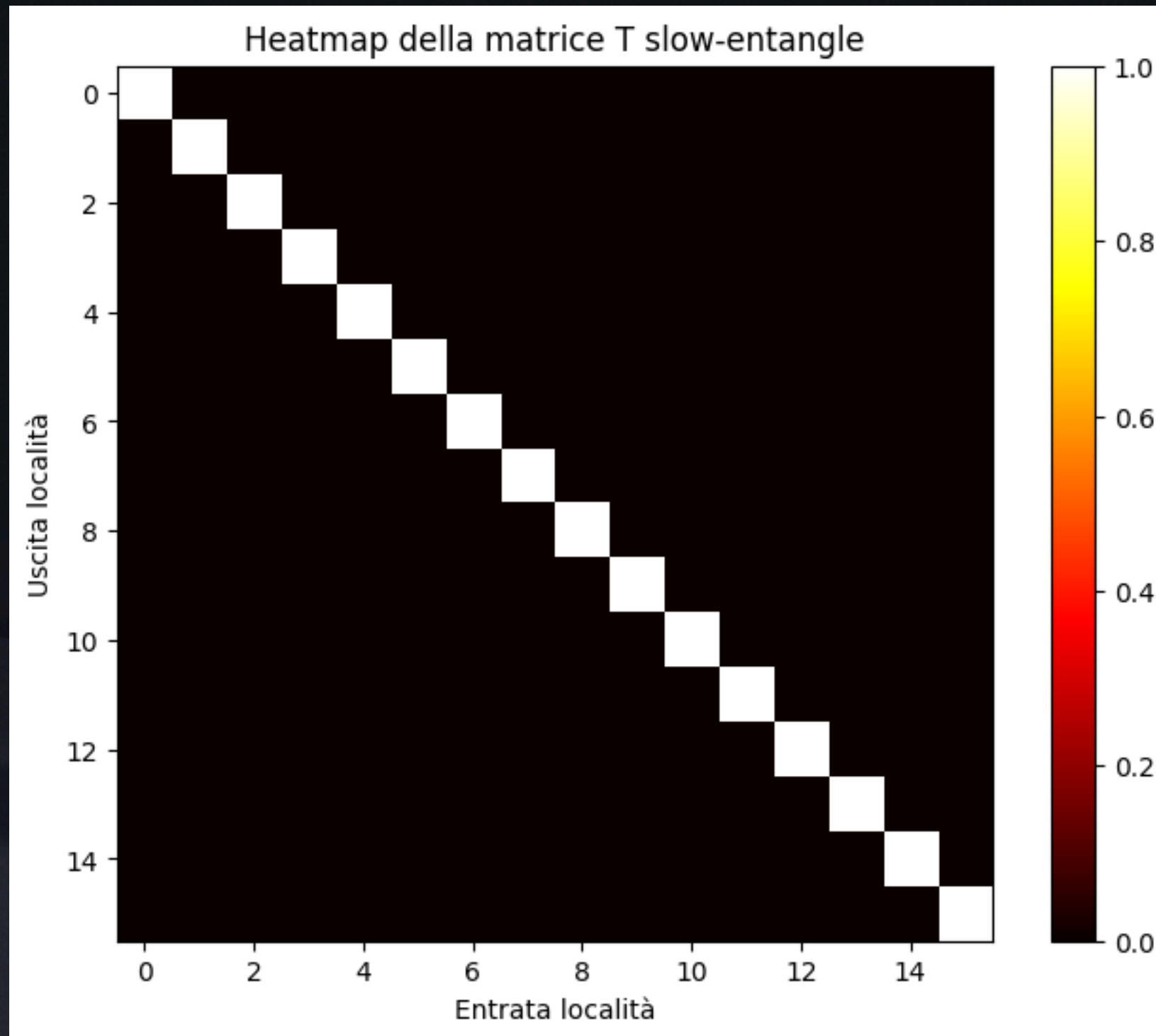
Slow entangling



Fast Entangling

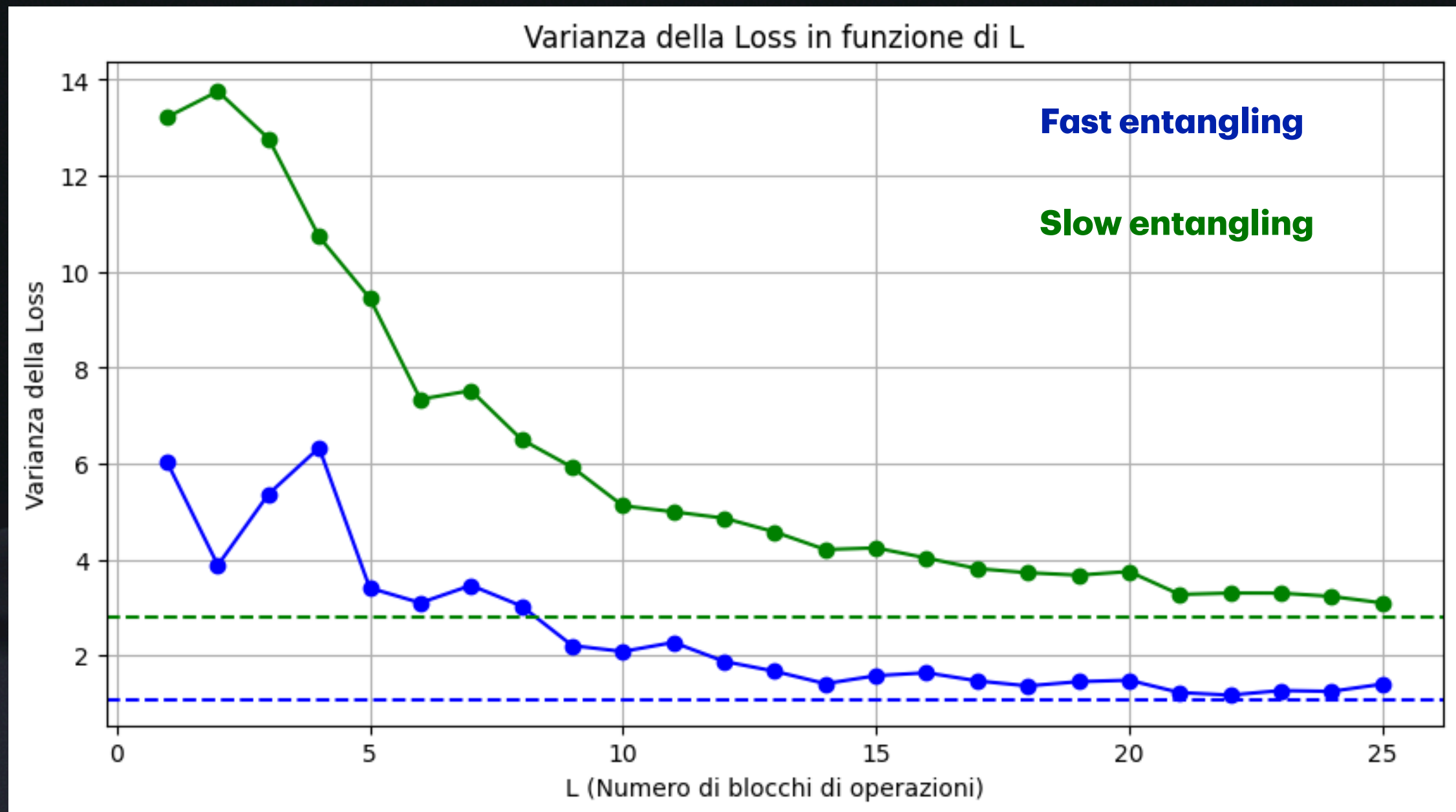


Slow Entangling

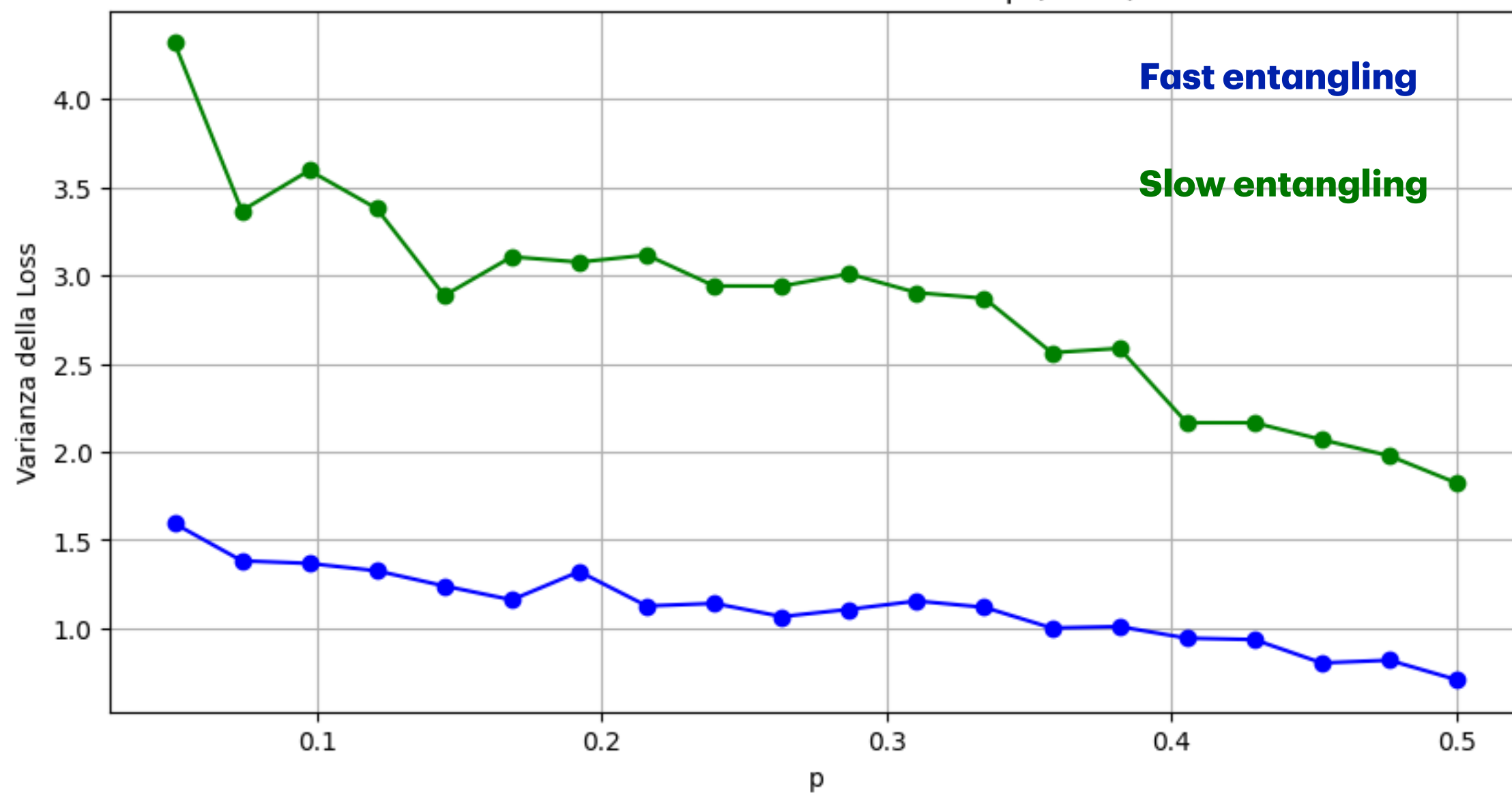


Varianza ottenuta

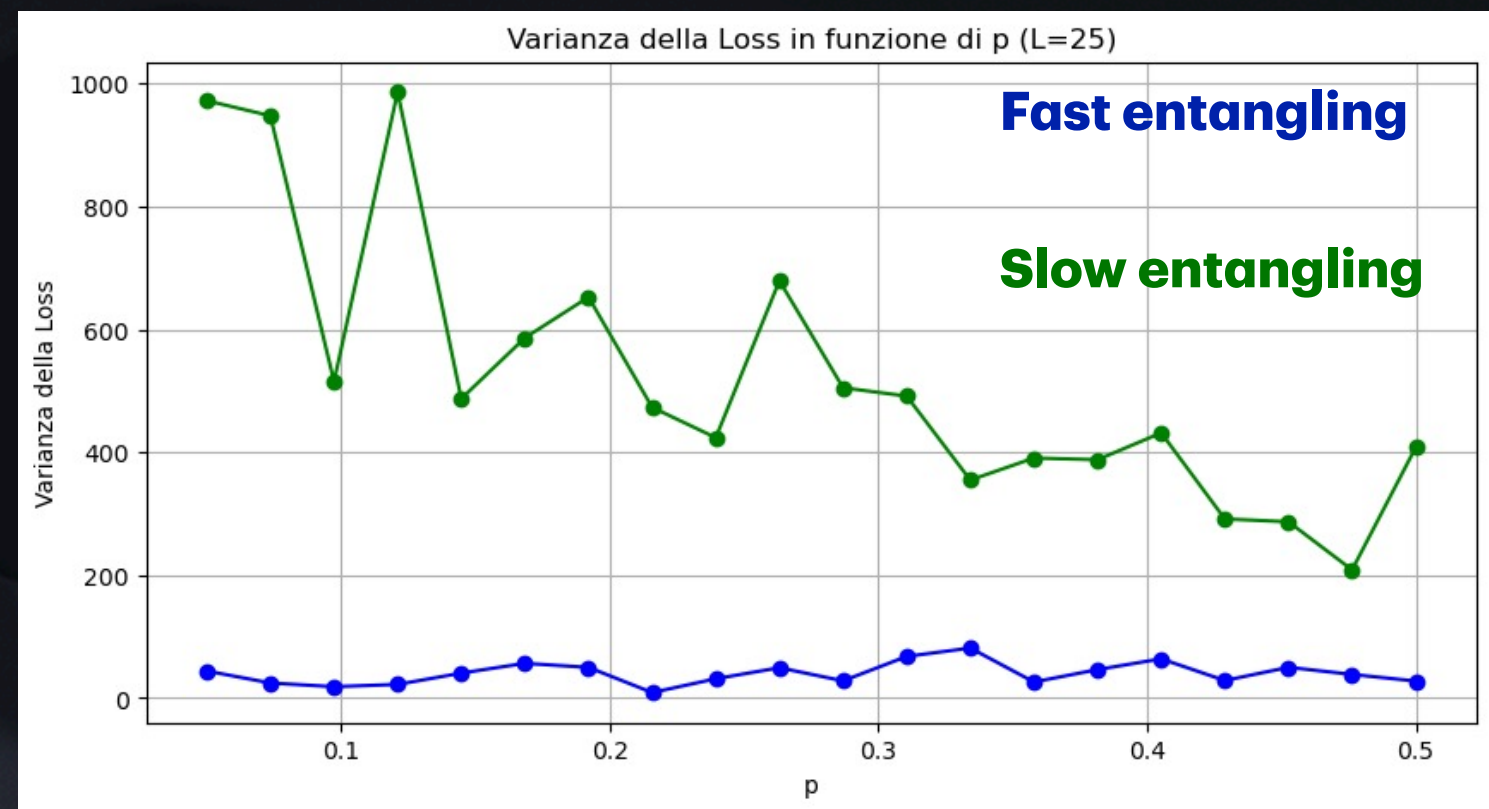
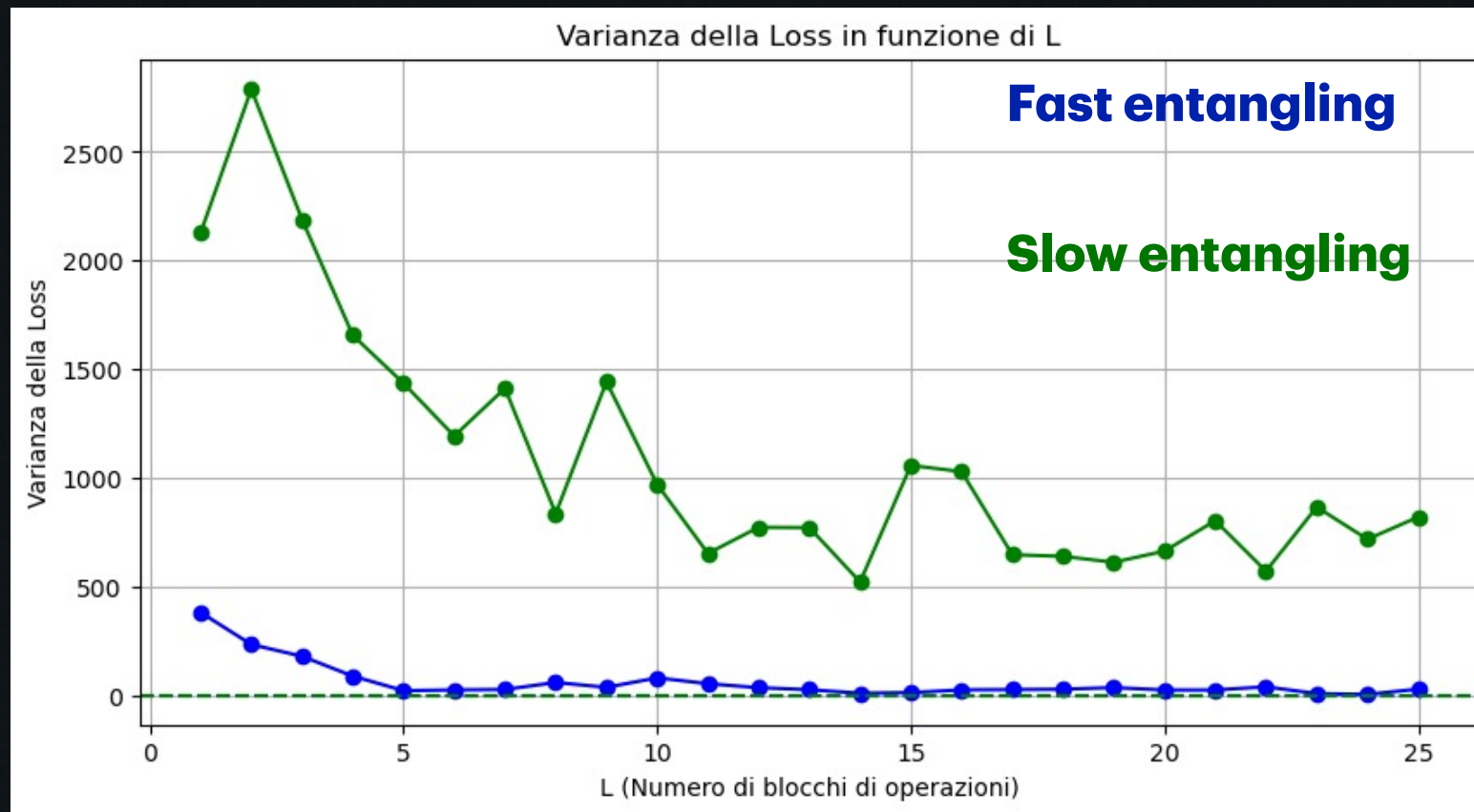
Fast e Slow



Varianza della Loss in funzione di p ($L=25$)

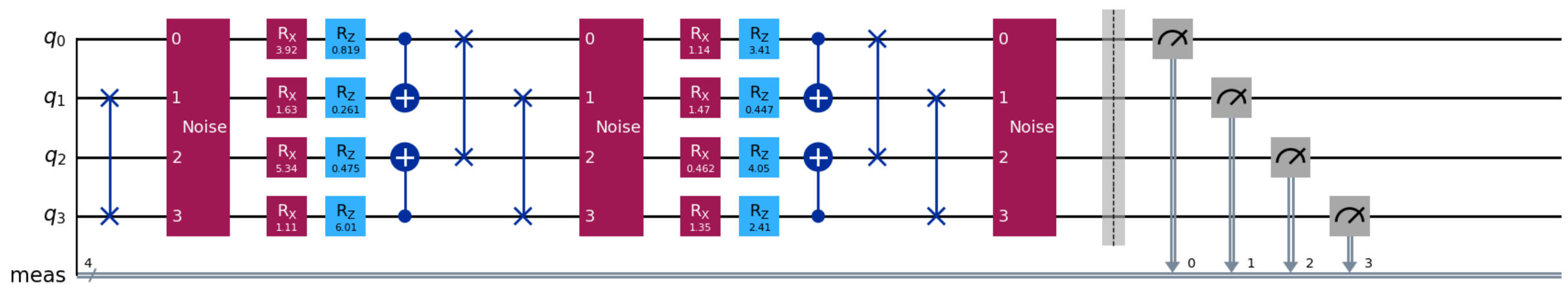
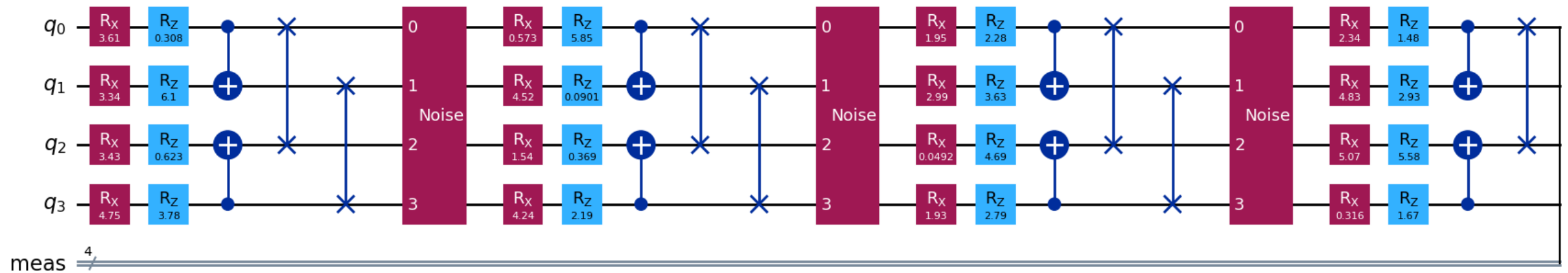


10 q-bit

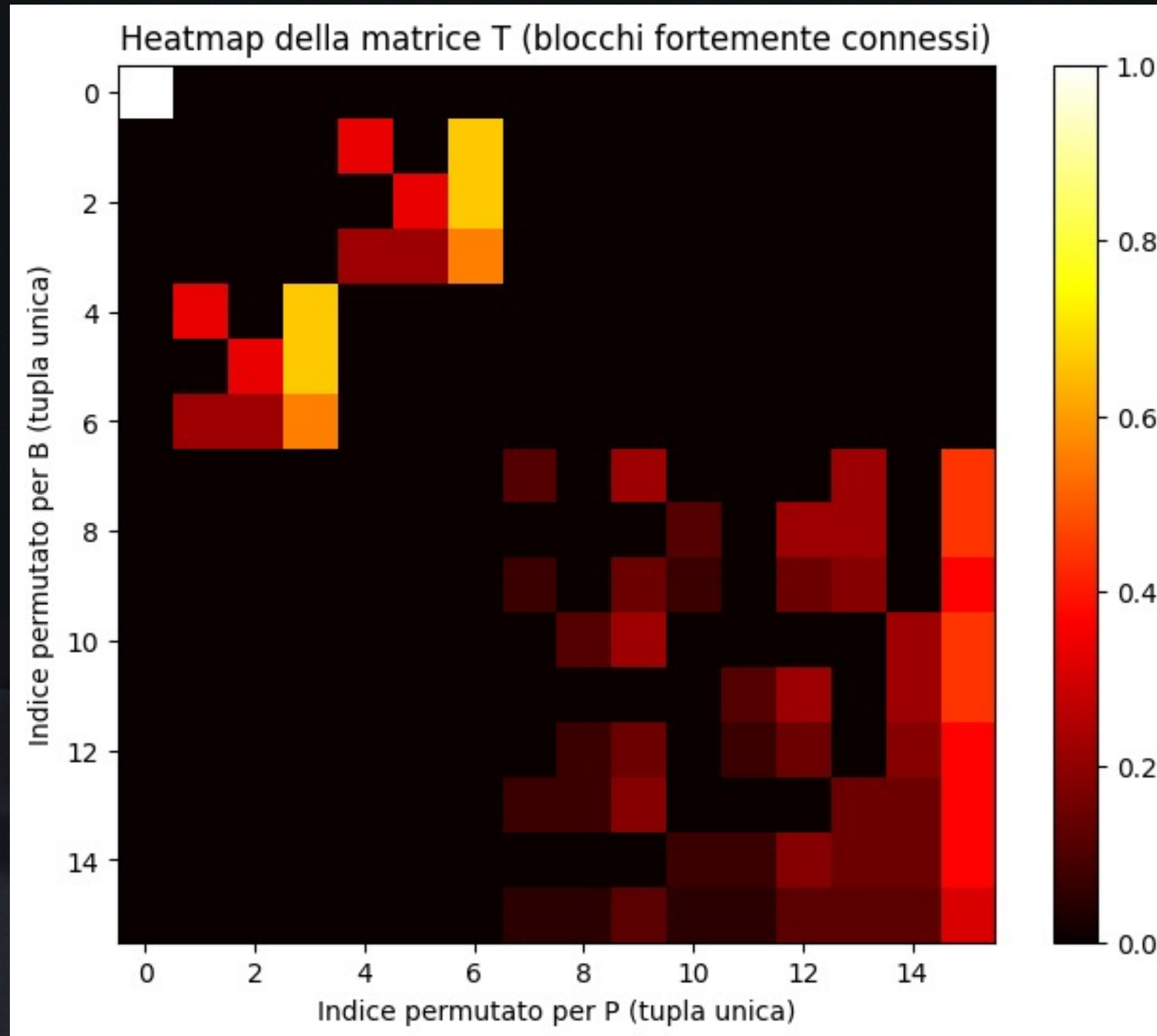


Circuito considerato

Intermediate Entangling

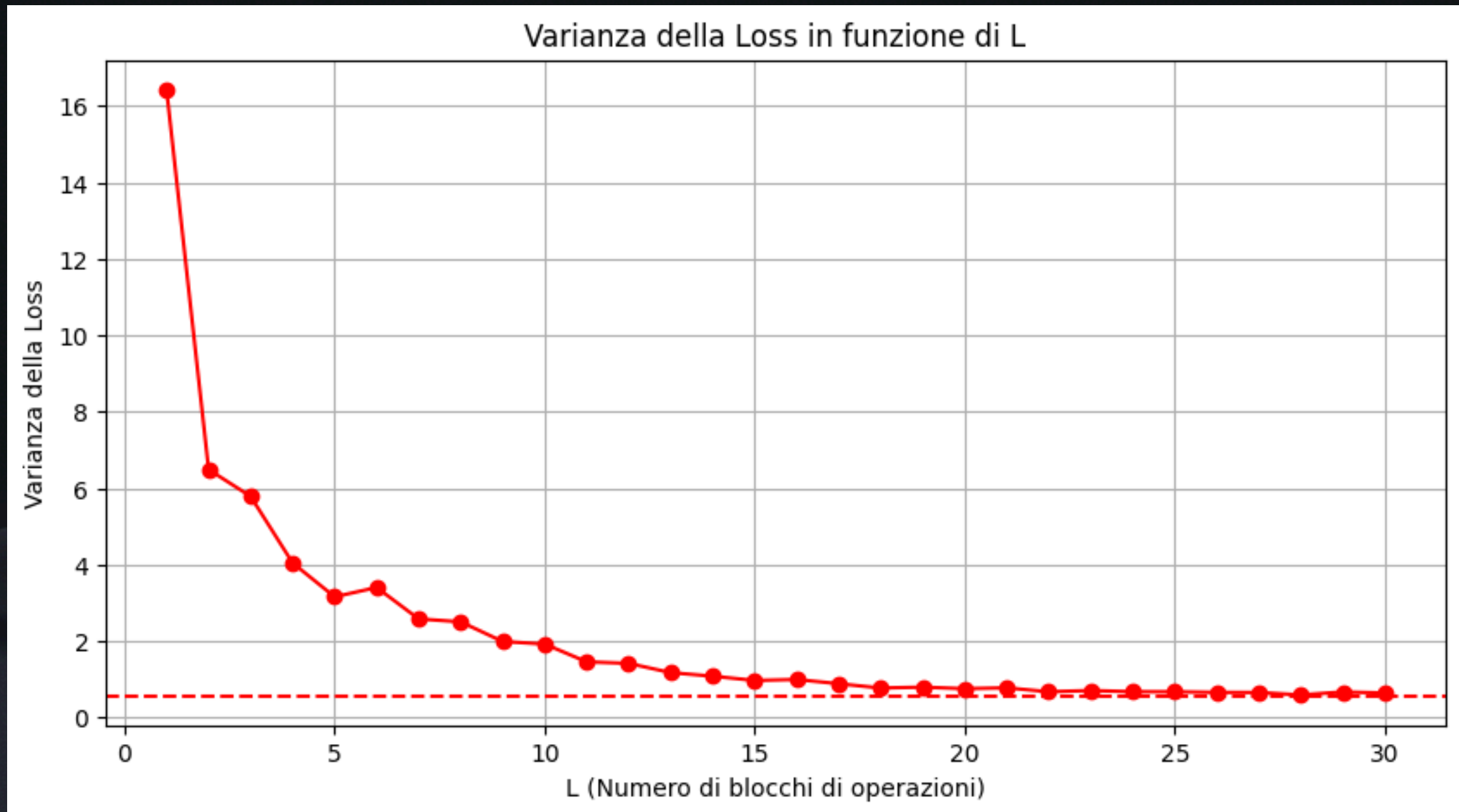


LTM



Varianza

Intermediate



Varianza della Loss in funzione di p ($L=30$)

